Braided-Symmetric Derivations of Algebras

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ABSTRACT
We study braided-symmetric derivations of an algebra over a field. These are maps that satisfy the Leibniz rule for the product symmetrized by an involution of $A \otimes A$. The identity involution gives the ordinary derivations when 2 is invertible. The swap gives the Jordan derivations. The negative of the identity gives all linear endomorphisms, and the negative of the swap gives the Lie derivations. We then twist the swap by an involutive automorphism or anti-automorphism α of A. The Leibniz defect of a map that commutes with α decides when it is a twisted derivation. For a commutative reduced algebra with 2 invertible, the twisted maps are exactly the derivations that commute with α, and on 2-torsion-free semiprime algebras the Jordan maps among them are exactly those derivations. On the algebra $\mathbb{F}_3[\varepsilon]/(\varepsilon^3)$ the twisted derivations (those commuting with α) and the α-derivations do not contain each other. Nilpotent elements show that reducedness is needed, characteristic 2 that the invertibility of 2 is needed, and a four-dimensional example that reducedness and commutativity cannot both be dropped.

KEYWORDS
Braided-symmetric derivations; twisted derivations; Jordan derivations; Leibniz defect

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1. Introduction

Throughout, K is a field and an algebra means a unital associative K-algebra. We take all maps between algebras to be K-linear. We say so when a result cited below holds more generally for additive maps on rings.

A *derivation* of an algebra A is a linear map $D\colon A \to A$ with $D(xy) = D(x)y + xD(y)$ for all $x, y \in A$. The notion was put on an abstract footing by Jacobson [12, Part I], who studied the derivations of an arbitrary nonassociative algebra. Hochschild [9, §§2 and 4] defined the cocycles of an associative algebra, and its 1-cocycles are the derivations. A *Lie derivation* is a linear map with $D([x, y]) = [D(x), y] + [x, D(y)]$ [14, p. 183]. Benkovič [2, §1] treats derivations, Jordan derivations and Lie derivations as f-derivations, for $f = x_1x_2$, $x_1 \circ x_2$ and $[x_1, x_2]$, so that one identity for a multilinear polynomial f covers all three. Beidar et al. [1, Theorem 4.3] studied maps that act as derivations on a fixed

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multilinear polynomial, earlier than Benkovič. A *Jordan derivation* of A is a linear map with $D(x \circ y) = D(x) \circ y + x \circ D(y)$ for all $x, y \in A$, where $x \circ y = xy + yx$. The sum $xy + yx$ is twice the Jordan product $\frac{1}{2}(xy + yx)$ of an associative algebra [4, Example 2.1]. Jordan, von Neumann and Wigner form a product ab from the two products $a \times b$ and $b \times a$ of an associative system, and call this quasi-multiplication [13, p. 64]. Herstein [7, p. 1104] takes the sum $ab+ba$ without the half. We follow Herstein and do not divide by 2, so that characteristic 2 is allowed. The results below need three notions for a ring R. We call R *prime* if $aRb = 0$ implies $a = 0$ or $b = 0$ [3, p. 1003]. We call R *semiprime* if $aRa = 0$ implies $a = 0$ [3, p. 1003]. We call R 2*-torsion-free* if $2a = 0$ implies $a = 0$ [3, p. 1003]. So every prime ring is semiprime. Herstein proved that every Jordan derivation of a prime ring of characteristic different from 2 is a derivation [7, Theorem 3.1]. Every derivation is a Jordan derivation, and the converse is false in general [3, p. 1003]. Cusack [6, Corollary 5] extended this result of Herstein to 2-torsion-free semiprime rings, and Brešar [3, Theorem 1] gave an independent proof.

Let A be an algebra over K with product μ. Let $\sigma\colon A \otimes A \to A \otimes A$ be a K-linear map with $\sigma^2 = \mathrm{id}$, for instance the swap $\tau(x \otimes y) = y \otimes x$. It gives a symmetrized product, as the Jordan product does [4, Example 2.1]

$$x \cdot_\sigma y = \mu(x \otimes y) + \mu(\sigma(x \otimes y)).$$

For $\sigma = \tau$ this is the Jordan product $x \circ y$. Put $\mu_\sigma = \mu + \mu \circ \sigma$. Since $\sigma^2 = \mathrm{id}$, we get $\mu_\sigma \circ \sigma = \mu_\sigma$, that is, the following triangle commutes.

$$\begin{array}{ccc} A \otimes A & \xrightarrow{\ \sigma\ } & A \otimes A \\ & \searrow^{\mu_\sigma} & \downarrow{\scriptstyle \mu_\sigma} \\ & & A \end{array}$$

Example 2.3 shows that a twisted version of this product can fail to be associative and to have a unit.

We call a linear map $D\colon A \to A$ a *braided-symmetric derivation* for σ if

$$D(x \cdot_\sigma y) = D(x) \cdot_\sigma y + x \cdot_\sigma D(y). \tag{1}$$

So (1) is the Leibniz rule for the symmetrized product. Other authors have studied *braided derivations*, which use a braiding operator directly [16, §2] [11, §4.1, Definition 14]. Remark 2.2 keeps the two notions apart. Our identity uses only μ, σ and D. We use the word *braided* by analogy with these braiding operators: no braided monoidal category is involved, and σ is any K-linear involution of $A \otimes A$.

We now describe the main results. Our first step is a dictionary. Remark 2.4 shows that the identity gives exactly the derivations when 2 is invertible, the swap gives exactly the Jordan derivations, and the two sign changes give all linear endomorphisms and the Lie derivations.

Our second step twists the swap by an involutive automorphism or anti-automorphism α of A. For an involutive α we put $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$, and we write $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ for the maps of the class of Definition 2.1 that commute with α; we call them *twisted derivations*; the literature's twisted (σ-)derivations are a different notion, the α-derivations of Section 2. Proposition 3.2 describes the twisted derivations through the defect $E_D(x, y) = D(xy) - D(x)y - xD(y)$. Corollary 3.5 shows that, on 2-torsion-free semiprime algebras, the Jordan maps among the twisted derivations are the derivations

commuting with α.

Theorem 4.1 treats commutative reduced algebras and needs no Jordan hypothesis. Example 4.2 shows that the reduced hypothesis is needed, and Example 4.3 shows the same for the invertibility of 2. Proposition 4.4 shows that the twisted derivations and the α-derivations on one algebra do not contain each other.

The dictionary itself is not new. The four structure-free involutions give the products $2xy$, $xy+yx$, 0 and $xy-yx$, which in the notation $f(x,y)=\beta xy+\gamma yx$ of Benkovič [2, Remark 1.4] are the cases $\beta=2$, $\gamma=0$; $\beta=\gamma=1$; and $\beta=1$, $\gamma=-1$; that remark assigns a Lie derivation to $\gamma=-\beta$, a Jordan derivation to $\gamma=\beta$ and a derivation to $\gamma\neq\pm\beta$, under the regularity hypotheses stated there, and the degenerate zero product $\beta=\gamma=0$ lies outside its cases. The contribution of this paper is elsewhere: the criterion of Proposition 3.2, which turns each of the questions below into a linear condition; Theorem 4.1, which needs no Jordan hypothesis; Proposition 4.4, which shows that the two classes are incomparable; and Remark 4.6, which gives a counterexample when commutativity and reducedness are dropped together and reduces the remaining case.

We follow this order. Section 2 recalls the classical notions, defines braided-symmetric derivations and treats the four structure-free involutions, those that use nothing of A beyond its K-vector space, which collapse to two in characteristic 2 (and to one when $\dim_K A\leq 1$). Section 3 treats the twisted swap through the Leibniz defect. Section 4 treats reduced commutative algebras and compares the twisted derivations with the α-derivations. Most examples follow the result they illustrate.

2. Preliminaries and braided-symmetric derivations

This section recalls the classical notions we need, defines braided-symmetric derivations and identifies them for the four structure-free involutions.

From now on, A is a K-algebra with unit 1. We write $[x,y]=xy-yx$ for the commutator, $x\circ y=xy+yx$ for the Jordan product, and $Z(A)$ for the centre. The centre is the set of elements z with $zx=xz$ for all $x\in A$. We call an element x *nilpotent* if $x^n=0$ for some $n\geq 1$. An algebra is *reduced* if its only nilpotent element is zero; for commutative algebras this is the notion of Chambert-Loir [5, 1.4.12], and Remark 4.6 uses it for an algebra that is not commutative.

A derivation is defined in the introduction. Every derivation satisfies $D(1)=0$, since $D(1)=D(1\cdot 1)=2D(1)$. For $g\in A$, the map $\mathrm{ad}_g(x)=[g,x]$ is an *inner derivation*; with the convention $[g,x]=gx-xg$ of this paper, the principal derivations $x\mapsto xm-mx$ of Weibel [17, §9.2] are the maps ad_{-m}. We write $\mathrm{Der}(A)$ and $\mathrm{Inn}(A)$ for these two spaces.

We add a word on the characteristic for Jordan derivations. Herstein [7, p. 1104] defines a *Jordan derivation* as a derivation of the Jordan ring of A, that is by $D(x\circ y)=D(x)\circ y+x\circ D(y)$, and records that for rings of characteristic different from 2 it can be defined by the square identity $D(x^2)=D(x)x+xD(x)$. We write $\mathrm{JDer}(A)$ for the space of linear maps with $D(x\circ y)=D(x)\circ y+x\circ D(y)$ for all x,y. Replacing x by $x+y$ in the square identity gives the condition $D(x\circ y)=D(x)\circ y+x\circ D(y)$ in every characteristic. Setting $y=x$ in this condition gives $2D(x^2)=2(D(x)x+xD(x))$. So the two conditions agree when 2 is invertible in K, and also when A is 2-torsion-free, where the factor 2 in the displayed identity can be cancelled; we use the $\circ$ form throughout. When $\mathrm{char}\,K=2$ and A is commutative, $x\circ y=0$, and every linear map satisfies the $\circ$ condition.

For an automorphism α of A, an α-*derivation* satisfies $D(xy)=D(x)\alpha(y)+\alpha(x)D(y)$

[10, §1]. We write $\mathrm{Der}_\alpha(A)$ for this space.

We now compare the Jordan derivations with the derivations. Every derivation is a Jordan derivation [3, p. 1003], so $\mathrm{Der}(A) \subseteq \mathrm{JDer}(A)$.

Definition 2.1. Let A be a K-algebra and let σ be a K-linear involution of $A \otimes A$. A *braided-symmetric derivation* for σ is a K-linear map $D\colon A \to A$ that satisfies identity (1), that is, $D(x \cdot_\sigma y) = D(x) \cdot_\sigma y + x \cdot_\sigma D(y)$ for all $x, y \in A$. We write $\mathrm{BSDer}_\sigma(A)$ for the set of braided-symmetric derivations for σ. This set is a K-subspace of $\mathrm{End}_K(A)$, since the condition is linear in D. For an automorphism or an anti-automorphism α of A (both are recalled in Section 3), we write $\mathrm{BSDer}_\sigma(A)^\alpha$ for the subset of maps D with $D\alpha = \alpha D$.

Identity (1) says that the following square commutes.

$$\begin{array}{ccc} A \otimes A & \xrightarrow{\mu_\sigma} & A \\ {\scriptstyle D\otimes 1 + 1 \otimes D}\downarrow & & \downarrow{\scriptstyle D} \\ A \otimes A & \xrightarrow[\mu_\sigma]{} & A \end{array}$$

The name is close to an older one, and the next remark separates the two.

Remark 2.2. Braided derivations [16, §2] [11, §4.1, Definition 14] form a different notion from braided-symmetric derivations, although both use a braiding. Definition 2.1 asks instead for the ordinary Leibniz rule for the new product $\mu + \mu \circ \sigma$.

Example 2.3 shows that the product $\cdot_\sigma$ can fail to be associative and unital, even when A is commutative.

Example 2.3. Let $A = K \times K$ with componentwise operations, and let $\alpha(a, b) = (b, a)$. Then α is an automorphism with $\alpha^2 = \mathrm{id}$, and $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$ is an involution of $A \otimes A$. Since A is commutative, $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$. For $x = (x_1, x_2)$ and $y = (y_1, y_2)$ this gives $x \cdot_{\sigma_\alpha} y = (x_1y_1 + x_2y_2)(1, 1)$. Put $p = (1, 0)$ and $q = (0, 1)$. Then $(p \cdot_{\sigma_\alpha} p) \cdot_{\sigma_\alpha} q = (1, 1) \cdot_{\sigma_\alpha} q = (1, 1)$, while $p \cdot_{\sigma_\alpha} (p \cdot_{\sigma_\alpha} q) = p \cdot_{\sigma_\alpha} 0 = 0$. So $\cdot_{\sigma_\alpha}$ is not associative. Every product $u \cdot_{\sigma_\alpha} y$ lies in $K(1, 1)$, so $u \cdot_{\sigma_\alpha} p \neq p$ for every u, and $\cdot_{\sigma_\alpha}$ has no unit.

The next remark records what Definition 2.1 says for the four structure-free involutions.

Remark 2.4. Let A be a K-algebra and let σ be a K-linear involution of $A \otimes A$. Identity (1) for σ is the Leibniz rule for the product $\cdot_\sigma$. So $\mathrm{BSDer}_\sigma(A)$ is the set of derivations of the nonassociative algebra $(A, \cdot_\sigma)$, in the sense of Jacobson [12, Part I].

For the identity, $x \cdot_{\mathrm{id}} y = 2xy$, so the identity reads

$$2D(xy) = 2\big(D(x)y + xD(y)\big).$$

If $\mathrm{char}\, K \neq 2$, we divide by 2 and get $\mathrm{BSDer}_{\mathrm{id}}(A) = \mathrm{Der}(A)$. If $\mathrm{char}\, K = 2$, then $x \cdot_{\mathrm{id}} y = 0$, so $\mathrm{BSDer}_{\mathrm{id}}(A)$ contains every linear map. If $A \neq 0$, this is larger than $\mathrm{Der}(A)$: the linear map with $D(1) = 1$ and $D = 0$ on a complement of $K1$ is not a derivation, since a derivation satisfies $D(1) = D(1 \cdot 1) = 2D(1) = 0$ in characteristic 2.

For the swap τ we get $x \cdot_\tau y = xy + yx = x \circ y$. So the defining identity is the $\circ$ form of the identity for a Jordan derivation recalled above. Hence $\mathrm{BSDer}_\tau(A) = \mathrm{JDer}(A)$ in every characteristic.

Every derivation is a Jordan derivation [3, p. 1003]. So in every characteristic

$$\mathrm{Der}(A) \subseteq \mathrm{BSDer}_\tau(A) = \mathrm{JDer}(A).$$

When $\operatorname{char} K \neq 2$, this gives $\mathrm{BSDer}_{\mathrm{id}}(A) \subseteq \mathrm{BSDer}_\tau(A)$.

The involution $-\mathrm{id}$ gives all linear endomorphisms, and $-\tau$ gives the Lie derivations. Their products are

$$x \cdot_{-\mathrm{id}} y = xy - xy = 0, \qquad x \cdot_{-\tau} y = xy - yx = [x, y].$$

So $\mathrm{BSDer}_{-\mathrm{id}}(A) = \mathrm{End}_K(A)$, since every linear map is a derivation of the zero product, and $\mathrm{BSDer}_{-\tau}(A)$ is the space of Lie derivations of A [14, p. 183]. In characteristic 2 the four involutions collapse to two, $-\mathrm{id} = \mathrm{id}$ and $-\tau = \tau$; when $\dim_K A \leq 1$ they collapse further, since $\tau = \mathrm{id}$ on $K \otimes K$. Remark 2.5 shows that in characteristic different from 2 these are the only natural involutions of the functor $V \mapsto V \otimes V$.

Remark 2.5. The four entries of Remark 2.4 are the involutions of $A \otimes A$ given by natural transformations of the functor $V \mapsto V \otimes V$ on K-vector spaces, that is, natural for all K-linear maps and not only for algebra maps; the natural endomorphisms of the functor are the linear combinations of id and τ, for three reasons: such an endomorphism is determined by its value on K^2, since every element of $V \otimes V$ is a sum of elements of the subspaces $j \otimes j(K^2 \otimes K^2)$ over the linear maps $j\colon K^2 \to V$, while for $\dim V \leq 1$ the assertion is immediate because $\mathrm{id} = \tau$ on $V \otimes V$; naturality in the maps $f\colon K^2 \to K^2$ makes that value an endomorphism of $K^2 \otimes K^2$ commuting with every $f \otimes f$, and those are the linear combinations of id and τ by a direct computation with the 2×2 matrices; and id and τ are natural. In characteristic different from 2 an involution among them satisfies $\lambda\mu = 0$ and $\lambda^2 + \mu^2 = 1$, so it is one of the four. Every idempotent P of $\mathrm{End}_K(A \otimes A)$ gives the involution $\mathrm{id} - 2P$, and when 2 is invertible every involution arises this way, from $P = \frac{1}{2}(\mathrm{id} - \sigma)$; the four structure-free idempotents are 0, id and $\frac{1}{2}(\mathrm{id} \pm \tau)$, whose reflections are the four entries of Remark 2.4, while, when $2 \neq 0$ and $\dim_K A \geq 2$, an idempotent that uses the multiplication, such as $a \otimes b \mapsto ab \otimes 1$, gives an involution that is none of them. In characteristic 2 the maps $\lambda\,\mathrm{id} + (1 + \lambda)\tau$, for $\lambda \in K$, are involutions as well, and they are the only natural ones; the product one of them defines is the scalar multiple $(1 + \lambda)(x \circ y)$ of the Jordan product, so they add no class to the dictionary, and the value $\lambda = 1$ gives the zero product. Involutions that use structure of A give other classes: for invertible T the map $(T \otimes T^{-1}) \circ \tau$ is an involution, and on $\mathbb{F}_5[\varepsilon]/(\varepsilon^2)$ with $T(1) = 1$ and $T(\varepsilon) = 1 + \varepsilon$ a direct computation on the two basis elements gives $\mathrm{BSDer}_\sigma(A) = \{0\}$, which is none of the four entries of Remark 2.4.

3. Twisted involutions and the Leibniz defect

Remark 2.4 treated the four structure-free involutions. This section twists the swap by an involutive automorphism or anti-automorphism α of A and treats it with one tool, the Leibniz defect.

We recall the notions used here. An *automorphism* of A is a bijective K-linear map $\alpha\colon A \to A$ with $\alpha(xy) = \alpha(x)\alpha(y)$ and $\alpha(1) = 1$. An *anti-automorphism* of A is a bijective K-linear map $\alpha\colon A \to A$ with $\alpha(xy) = \alpha(y)\alpha(x)$ and $\alpha(1) = 1$. By $(xy)^T = y^T x^T$, the transpose is an anti-automorphism of $M_n(K)$. In both cases α is *involutive* if $\alpha^2 = \mathrm{id}$. An involutive anti-automorphism is an involution of the ring A in the sense

of Herstein [8, §1.2], that is, a map $*$ with $a^{**} = a$, $(a+b)^* = a^* + b^*$ and $(ab)^* = b^*a^*$.

We need $\alpha^2 = \mathrm{id}$ for the following reason. Applying σ_α below twice gives

$$x \otimes y \mapsto \alpha^2(x) \otimes \alpha^2(y),$$

so $\sigma_\alpha^2 = \alpha^2 \otimes \alpha^2$, and σ_α is an involution of $A \otimes A$ exactly when $\alpha^2 = \mathrm{id}$: if $\alpha^2 \otimes \alpha^2 = \mathrm{id}$, then $\alpha^2 = \lambda\,\mathrm{id}$ with $\lambda^2 = 1$ (an endomorphism β with $\beta \otimes \beta = \mathrm{id}$ is a scalar multiple of the identity: choosing y with $\beta(y) \neq 0$, the identity $\beta(x) \otimes \beta(y) = x \otimes y$ forces $\beta(x) \in Kx$ for every x, with a scalar independent of x), and $\alpha^2(1) = 1$ gives $\lambda = 1$. When 2 is invertible, an involutive α splits $A = A_+ \oplus A_-$, where $A_\pm = \{x \in A : \alpha(x) = \pm x\}$. Indeed, for $x \in A$,

$$x = \tfrac{1}{2}\big(x + \alpha(x)\big) + \tfrac{1}{2}\big(x - \alpha(x)\big).$$

We call the elements of A_+ *symmetric* and the elements of A_- *skew*. The two sets are defined by the equations $\alpha(x) = \pm x$ and make sense without the hypothesis on 2; the invertibility of 2 is used only for the splitting: to see that they span A and that $A_+ \cap A_- = 0$.

Let α be involutive, of either kind. Then $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$ is a K-linear involution of $A \otimes A$. It is the composite of the swap and $\alpha \otimes \alpha$, as the following diagram shows.

$$A \otimes A \xrightarrow{\ \tau\ } A \otimes A \xrightarrow{\ \alpha\otimes\alpha\ } A \otimes A, \qquad A \otimes A \xrightarrow{\ \sigma_\alpha\ } A \otimes A$$

So σ_α defines a symmetrized product as in Section 2. The product it defines is

$$x \cdot_{\sigma_\alpha} y = xy + \alpha(y)\alpha(x). \tag{2}$$

For $\alpha = \mathrm{id}$ the product μ_{σ_α} is the Jordan product. If α is an automorphism, then $\alpha(y)\alpha(x) = \alpha(yx)$. If α is an anti-automorphism, then $\alpha(y)\alpha(x) = \alpha(xy)$.

Definition 3.1. The *Leibniz defect* of a linear map $D\colon A \to A$ is the bilinear map $E_D\colon A \times A \to A$ given, for $x, y \in A$, by

$$E_D(x, y) = D(xy) - D(x)y - xD(y).$$

The name is ours, and the map is classical: Cusack [6, §2] writes it $d(a, b) = D(ab) - aDb - (Da)b$ and studies Jordan derivations through it. We recall the Hochschild complex of A with coefficients in A [17, §9.1]. An *n-cochain* is a K-multilinear map from A^n to A. So a 0-cochain is an element of A, and a 1-cochain is a linear map $D\colon A \to A$. The *coboundary* δ sends each n-cochain to an $(n+1)$-cochain, and $\delta \circ \delta = 0$ [17, §9.1]. An *n-cocycle* is an n-cochain f with $\delta f = 0$. The n-th *Hochschild cohomology* $HH^n(A, A)$ is the space of n-cocycles modulo the images δg of the $(n-1)$-cochains g [17, §9.1].

On a 1-cochain D the coboundary is $(\delta D)(x, y) = xD(y) - D(xy) + D(x)y$ [9, §2, Definition 2.1]. So $E_D = -\delta D$. Thus D is a derivation exactly when $E_D = 0$, so derivations are the 1-cocycles. Since $\delta \circ \delta = 0$, the defect is a 2-cocycle: for all $a, b, c \in A$,

$$aE_D(b, c) - E_D(ab, c) + E_D(a, bc) - E_D(a, b)c = 0. \tag{3}$$

This says that E_D is a cocycle of the Hochschild complex [17, §9.1].

The defect also describes the other classes of Section 2. Expanding $D(x \circ y)$ shows that D is a Jordan derivation exactly when $E_D(x,y) + E_D(y,x) = 0$ for all x, y.

Proposition 3.2. *Let A be a unital associative K-algebra, let α be an involutive automorphism or anti-automorphism of A, and let $D\colon A \to A$ be a linear map with $D\alpha = \alpha D$.*

(i) If α is an automorphism, then $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $E_D(x,y) + \alpha(E_D(y,x)) = 0$ for all $x, y \in A$.

(ii) If α is an anti-automorphism, then $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $E_D(x,y) + \alpha(E_D(x,y)) = 0$ for all $x, y \in A$.

In both cases every derivation that commutes with α lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$.

Proof. Write $E = E_D$. By Remark 2.4, $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ means $D(x \cdot_{\sigma_\alpha} y) = D(x) \cdot_{\sigma_\alpha} y + x \cdot_{\sigma_\alpha} D(y)$ for all x, y. We compute the difference of the two sides in each case.

For (i), the product (2) is $x \cdot_{\sigma_\alpha} y = xy + \alpha(yx)$. Since $D\alpha = \alpha D$, the left side is $D(xy) + \alpha(D(yx))$. The right side is $D(x)y + \alpha(yD(x)) + xD(y) + \alpha(D(y)x)$. We group the terms without α and the terms with α. The first group gives $D(xy) - D(x)y - xD(y) = E(x,y)$. Since α is linear, the second group gives $\alpha\big(D(yx) - D(y)x - yD(x)\big) = \alpha(E(y,x))$. So the left side minus the right side is $E(x,y) + \alpha(E(y,x))$, which proves (i).

The proof of (ii) follows the same steps as the proof of (i). Here the product (2) is $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$. Since $D\alpha = \alpha D$, the left side is

$$D(xy) + \alpha(D(xy)),$$

and the right side is

$$D(x)y + \alpha(D(x)y) + xD(y) + \alpha(xD(y)).$$

The same grouping gives the difference

$$E(x,y) + \alpha(E(x,y)),$$

which proves (ii).

If D is a derivation, then $E = 0$, and both conditions hold. □

Without the hypothesis $D\alpha = \alpha D$ the same computation gives the residual identities

$$D(x \cdot_{\sigma_\alpha} y) - D(x) \cdot_{\sigma_\alpha} y - x \cdot_{\sigma_\alpha} D(y) = E_D(x,y) + \alpha(E_D(y,x)) + (D\alpha - \alpha D)(yx)$$

in case (i), and the same with $\alpha(E_D(x,y))$ and $(D\alpha - \alpha D)(xy)$ in case (ii). So the hypothesis $D\alpha = \alpha D$ removes exactly the last term: with it, the two remaining terms vanish exactly when D lies in the class, by Proposition 3.2. Without it they no longer characterize the class, and when A is unital they say something else: if they vanish, then the difference of the two sides of the σ_α-identity is $(D\alpha - \alpha D)(yx)$ in case (i) and $(D\alpha - \alpha D)(xy)$ in case (ii); since the products yx, and likewise xy, span a unital A, D lies in the class exactly when $D\alpha = \alpha D$.

The computation uses neither the unitality of A nor the involutivity of α; involutivity is needed only for σ_α to be an involution.

We now read the two conditions; the second needs no hypothesis, and for the first we assume that 2 is invertible.

In case (ii), the equation $E_D + \alpha(E_D) = 0$ says $\alpha(E_D) = -E_D$. So E_D takes values in A_-, the skew elements of α. In case (i), we write $E_D = E_{\mathrm{s}} + E_{\mathrm{a}}$. Here $E_{\mathrm{s}}(x,y) = \frac{1}{2}(E_D(x,y) + E_D(y,x))$ is symmetric. And $E_{\mathrm{a}}(x,y) = \frac{1}{2}(E_D(x,y) - E_D(y,x))$ is antisymmetric.

We swap x and y in the condition of (i). Then we take the sum and the difference of the two equations. So the condition holds exactly when E_{s} takes values in A_- and E_{a} takes values in A_+. Now let $\alpha = \mathrm{id}$. Then $A_- = 0$ and $A_+ = A$, so the condition says $E_{\mathrm{s}} = 0$. This is the Jordan condition, so Proposition 3.2(i) recovers the swap case of Remark 2.4.

Remark 3.3. Let α be an involutive automorphism or anti-automorphism of A, and let $D \in \mathrm{Der}(A)$. Then $E_D = 0$. By (2), the two sides of the σ_α-identity differ by

$$D(\alpha(yx)) - \alpha(D(yx)),$$

and for an anti-automorphism the same computation with $\alpha(xy)$ in place of $\alpha(yx)$ applies. Since A is unital, the products yx span A. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $D\alpha = \alpha D$.

Next we look at the inner derivations ad_g in this class. The condition $\mathrm{ad}_g\,\alpha = \alpha\,\mathrm{ad}_g$ means

$$[g - \alpha(g), \alpha(x)] = 0 \qquad \text{for all } x.$$

Since α is bijective, this holds if and only if $g - \alpha(g) \in Z(A)$. If α is an anti-automorphism instead, the same computation with $\alpha(x)\alpha(g)$ in place of $\alpha(xg)$ gives $[g + \alpha(g), \alpha(x)] = 0$ for all x, that is, $g + \alpha(g) \in Z(A)$. In the rest of this remark α is an automorphism. Let $\mathrm{Inn}_\alpha(A)$ be the space of the maps ad_g with $g - \alpha(g) \in Z(A)$. These are the inner derivations in $\mathrm{BSDer}_{\sigma_\alpha}(A)$, so $\mathrm{Inn}_\alpha(A) = \mathrm{Inn}(A) \cap \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. The quotient $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha / \mathrm{Inn}_\alpha(A)$ is the space of *twisted outer derivations*.

Cusack [6, Corollary 5] and Brešar [3, Theorem 1] proved that a Jordan derivation of a 2-torsion-free semiprime ring is a derivation.

Theorem 3.4 ([3, Theorem 1])**.** *Let R be a 2-torsion-free semiprime ring. Then every Jordan derivation $D\colon R \to R$ is a derivation.*

The next corollary applies Theorem 3.4 to the twisted derivations.

Corollary 3.5. *Let A be a 2-torsion-free semiprime K-algebra with an involutive K-algebra automorphism or anti-automorphism α. Then* $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \cap \mathrm{JDer}(A) = \mathrm{Der}(A)^\alpha$*, where* $\mathrm{Der}(A)^\alpha$ *is the set of derivations commuting with α (not to be confused with the space* $\mathrm{Der}_\alpha(A)$ *of α-derivations of Section 2, which is the notion the literature calls a twisted derivation).*

Proof. Every derivation is a Jordan derivation by Remark 2.4. With Proposition 3.2, this gives $\mathrm{Der}(A)^\alpha \subseteq \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \cap \mathrm{JDer}(A)$. Conversely, every Jordan derivation of A is a derivation by Theorem 3.4, which applies since a K-linear map is additive; the $\circ$ form of the Jordan condition and Herstein's x^2 form agree by the linearization of Section 2, which uses the 2-torsion-freeness assumed here. So a map in the intersection is a derivation that commutes with α. □

4. Reduced commutative algebras and twisted derivations

For commutative algebras the Jordan hypothesis in Corollary 3.5 can be dropped. Recall from Section 2 that a reduced commutative algebra has no nonzero nilpotent element.

Theorem 4.1. *Let A be a commutative reduced K-algebra with 2 invertible in K, and let α be an involutive automorphism of A. Then* $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha = \mathrm{Der}(A)^\alpha$.

Proof. Every derivation commuting with α lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 3.2. Conversely, let $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. Write $E = E_D$. Since A is commutative, E is symmetric. So E takes values in A_- by Proposition 3.2(i).

Since D commutes with α, it maps each of A_+ and A_- into itself; and since α is multiplicative, $A_iA_j \subseteq A_{i+j}$. So if x and y lie in the same part, then xy, $D(x)y$ and $xD(y)$ all lie in A_+, and $E(x,y)$ lies in

$$A_+ \cap A_- = 0.$$

Now let $x \in A_+$ and $y \in A_-$, and put $z = E(x,y)$. Then z lies in A_-, and so do xy, zx and xz, while zy lies in A_+. Apply the cocycle identity (3) to $(a,b,c) = (z,x,y)$:

$$zE(x,y) - E(zx,y) + E(z,xy) - E(z,x)y = 0.$$

The pairs (zx,y) and (z,xy) lie in A_-, so E vanishes on them, and we get $z^2 = E(z,x)y$. Apply (3) to $(a,b,c) = (x,z,y)$:

$$xE(z,y) - E(xz,y) + E(x,zy) - E(x,z)y = 0.$$

The pairs (z,y) and (xz,y) lie in A_-, and the pair (x,zy) lies in A_+, so E vanishes on all three and $E(x,z)y = 0$. Since E is symmetric, $z^2 = E(z,x)y = E(x,z)y = 0$. So $z = 0$ because A is reduced. Since E is bilinear and symmetric and $A = A_+ \oplus A_-$, we get $E = 0$. Hence D is a derivation. The proof uses the commutativity of A only to make E_D symmetric, and reducedness only to pass from $z \in A_-$ and $z^2 = 0$ to $z = 0$. It therefore gives two stronger statements. First, "reduced" may be replaced by "no nonzero $z \in A_-$ satisfies $z^2 = 0$", a condition that is empty when $\alpha = \mathrm{id}$. Second, commutativity is used nowhere else in the argument: for any unital K-algebra A with 2 invertible and any involutive automorphism α, if $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ has symmetric defect E_D and no nonzero $z \in A_-$ satisfies $z^2 = 0$, then D is a derivation. Remark 4.6 shows that the conclusion fails when commutativity and reducedness are dropped together. □

The theorem is not vacuous. Let $A = K[t]$ and $\alpha(t) = -t$, with $2 \neq 0$. Then A is commutative and reduced, and $\mathrm{Der}(A)^\alpha$ is the space of the derivations $f(t)\,d/dt$ with $f(-t) = -f(t)$, which contains $t\,d/dt$, so the theorem asserts the non-trivial equality $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha = \{f\,d/dt : f(-t) = -f(t)\}$. Over a perfect field every finite-dimensional instance has $\mathrm{Der}(A)^\alpha = 0$: a finite-dimensional reduced commutative algebra is a product of finite separable field extensions of K [15, §3.5], and a separable extension has no nonzero K-derivation.

Example 4.2 shows that the reduced hypothesis cannot be dropped, and Example 4.3 shows the same for the invertibility of 2.

Example 4.2. Let $A = \mathbb{F}_5[u,v]/(u^2,v^2)$. It has basis $\{1,u,v,uv\}$. Let α fix u and send v to $-v$. Let D send uv to uv and the other three basis elements to 0. Then D commutes

with α. It satisfies the Leibniz rule for $x \cdot_{\sigma_\alpha} y$ on all basis pairs. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. But D is not a derivation, since

$$D(uv) = uv \qquad \text{while} \qquad D(u)v + uD(v) = 0.$$

A direct computation verifies the three statements: commutation with α is linear in D and is checked on a basis, and the Leibniz rule is linear in D and bilinear in the pair, so it too is checked on basis pairs; the failure of the derivation rule is read at the single pair (u, v). Here $z = E(u, v) = uv$ is a nonzero element with $z^2 = 0$.

The next example shows that the invertibility of 2 is needed in Theorem 4.1.

Example 4.3. Let $A = \mathbb{F}_2 \times \mathbb{F}_2$, which is commutative and reduced, and let α swap the two factors. Put $p = (1, 0)$, $q = (0, 1)$ and $e = (1, 1)$, and let D be the linear map with $D(p) = D(q) = e$, so that $D(e) = 0$. Then D commutes with α. On the basis pairs (p, p), (q, q) and (p, q), both sides of the Leibniz rule for $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$ are zero. Here $p \cdot_{\sigma_\alpha} p = e$, so the two sides vanish because $D(e) = 0$ and $2e = 0$, not because the products vanish. For a pair with e in one entry, $x \cdot_{\sigma_\alpha} e = x + \alpha(x)$ lies in $\{0, e\}$, so the left side vanishes because $D(e) = D(0) = 0$, and the right side is $D(x) \cdot_{\sigma_\alpha} e$, which vanishes because $D(x)$ lies in $\{0, e\}$ and $e \cdot_{\sigma_\alpha} e = 2e = 0$. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. But D is not a derivation: since $p^2 = p$, a derivation would satisfy $D(p) = D(p^2) = 2pD(p) = 0$, while $D(p) = e$.

Our second natural candidate for the twisted derivations is the space $\mathrm{Der}_\alpha(A)$ of α-derivations. It was defined for automorphisms in Section 2. It differs from $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ in both directions, and one algebra shows both failures.

Proposition 4.4. *Let $A = \mathbb{F}_3[\varepsilon]/(\varepsilon^3)$, and let α be the automorphism with $\alpha(\varepsilon) = -\varepsilon$. Then*

$$\mathrm{Der}_\alpha(A) \not\subseteq \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \qquad \textit{and} \qquad \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \not\subseteq \mathrm{Der}_\alpha(A).$$

Proof. The algebra A has basis $\{1, \varepsilon, \varepsilon^2\}$. The map α fixes 1 and ε^2 and sends ε to $-\varepsilon$. So it is an involutive automorphism with $A_+ = \mathbb{F}_3 1 + \mathbb{F}_3 \varepsilon^2$ and $A_- = \mathbb{F}_3 \varepsilon$. Since A is commutative, E_D is symmetric for every linear map D. Let D be a linear map with $D\alpha = \alpha D$. Since E_D is symmetric, the condition $E_D(x, y) + \alpha(E_D(y, x)) = 0$ of Proposition 3.2(i) reads $E_D(x, y) + \alpha(E_D(x, y)) = 0$, that is, E_D takes values in A_-; so D lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)$ exactly when this holds. We give one map for each non-inclusion. Both maps are diagonal on the basis, so both commute with α.

For the first non-inclusion, let $D(a + b\varepsilon + c\varepsilon^2) = b\varepsilon + c\varepsilon^2$. We check that D is an α-derivation on pairs of basis elements. Pairs that contain 1 give D of the other element of the pair on both sides, since $D(1) = 0$ and $\alpha(1) = 1$. For the pair $(\varepsilon, \varepsilon)$, the left side is $D(\varepsilon^2) = \varepsilon^2$. The right side is $D(\varepsilon)\alpha(\varepsilon) + \alpha(\varepsilon)D(\varepsilon) = -2\varepsilon^2 = \varepsilon^2$. The remaining pairs have product 0, and each term on the right side is a multiple of $\varepsilon^3 = 0$. So $D \in \mathrm{Der}_\alpha(A)$. On the other hand, $E_D(\varepsilon, \varepsilon) = D(\varepsilon^2) - 2\varepsilon D(\varepsilon) = \varepsilon^2 - 2\varepsilon^2 = 2\varepsilon^2$. This element is nonzero and lies in A_+, so it does not lie in A_-. Hence $D \notin \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$.

The second non-inclusion uses a derivation.

Let d be the linear map

$$d(a + b\varepsilon + c\varepsilon^2) = b\varepsilon + 2c\varepsilon^2.$$

It is a derivation, since $d(\varepsilon^i) = i\varepsilon^i$ and exponents add under multiplication. So $E_d = 0$, and $d \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. On the other hand,

$$d(\varepsilon^2) - d(\varepsilon)\alpha(\varepsilon) - \alpha(\varepsilon)d(\varepsilon) = 2\varepsilon^2 + 2\varepsilon^2 = \varepsilon^2 \neq 0.$$

Hence $d \notin \mathrm{Der}_\alpha(A)$. □

The two classes can also meet, and one of them can fail to be defined. Part (b) of the next example shows the first, part (a) the second.

Example 4.5. (a) Let $A = M_2(\mathbb{F}_3)$ and let α be the transpose, an involutive anti-automorphism. Then $\mathrm{Der}_\alpha(A)$ is not defined, since it needs an automorphism, so this part gives a twisted derivation with no α-derivation to compare with. Take $g = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, which is nonzero and skew for the transpose, and let ad_g be the inner derivation $x \mapsto [g, x]$. It commutes with α by the anti-automorphism case of Remark 3.3, since $g + \alpha(g) = 0$, and it has $E = 0$, so it lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 3.2.

(b) Now let $A = K[\varepsilon]/(\varepsilon^2)$ with $\alpha(\varepsilon) = -\varepsilon$. The map $D(a + b\varepsilon) = b\varepsilon$ is a derivation and an α-derivation, and it commutes with α. So it lies in both $\mathrm{Der}_\alpha(A)$ and $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 3.2. A direct computation gives that when $\mathrm{char}\, K \neq 2$ both classes are one-dimensional, spanned by $\varepsilon \mapsto \varepsilon$, so here they coincide; when $\mathrm{char}\, K = 2$ the map α is the identity, $\mathrm{BSDer}_{\sigma_\alpha}(A)$ is the whole of $\mathrm{End}_K(A)$, and $\mathrm{Der}_\alpha(A) = \mathrm{Der}(A)$ is two-dimensional.

Remark 4.6 (Reducedness and commutativity cannot both be dropped)**.** Theorem 4.1 fails when the hypotheses that A is commutative and that A is reduced are both dropped. Let A be the four-dimensional $\mathbb{F}_3$-algebra with basis $1, t, v, w$, in which 1 is a unit and

$$t^2 = 0, \qquad tv = vt = tw = wt = 0, \qquad v^2 = w^2 = wv = t, \qquad vw = 0,$$

and let α be the linear map with $\alpha(t) = t$, $\alpha(v) = -v$ and $\alpha(w) = -w$. Then A is associative, a check on the 64 triples of basis elements, and α is an involutive automorphism. The linear map D with $D(v) = v + w$ and $D(1) = D(t) = D(w) = 0$ commutes with α and satisfies the Leibniz rule for $\cdot_{\sigma_\alpha}$ on every pair of basis elements; since that rule is linear in D and bilinear in the pair, D lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. It is not a derivation, since $D(vw) = 0$ while $D(v)w + vD(w) = vw + w^2 = t \neq 0$. Solving the linear conditions on the basis pairs gives $\dim \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha = 3$ and $\dim \mathrm{Der}(A)^\alpha = 2$. The defect is anti-symmetric at (v, w), where $E_D(v, w) = -t$, and A_- does contain a nonzero square-zero element, $(v+w)^2 = 0$; this is consistent with the symmetric-defect statement at the end of the proof of Theorem 4.1. The algebra is neither commutative ($vw = 0$ while $wv = t$) nor reduced ($t \neq 0$ and $t^2 = 0$), so the example does not show that commutativity alone is needed, and we make no such claim. A witness that drops commutativity while keeping reducedness is not to be had in finite dimension over a finite field: the radical of a finite-dimensional algebra is nilpotent [15, §4.4], hence zero in a reduced algebra, so the algebra is semisimple; by Wedderburn's structure theorem it is then a product of matrix algebras over division rings [15, §3.5], and reducedness forces the matrix sizes to be 1, since $e_{12}^2 = 0$ in $M_n(D)$ for $n \geq 2$; so it is a product of division rings, each of which is a field when the base field is finite. The first candidate is therefore a non-commutative division algebra, such as the quaternions $\mathbb{H}$ over $\mathbb{R}$; there $\alpha = \mathrm{id}$ gives $\sigma_\alpha = \tau$, so the assertion becomes $\mathrm{JDer}(\mathbb{H}) = \mathrm{Der}(\mathbb{H})$, which is Theorem 3.4. The quaternions do not

provide a witness either. Every automorphism of $\mathbb{H}$ is inner [15, §12.6], so the involutive ones are the identity and the conjugations by pure imaginary quaternions; any two of the latter are conjugate, since $uqu^{-1} = q'$ for $u = 1 - q'q$, which is nonzero because $u\bar{u} = 2 - (qq' + q'q)$ and $qq' + q'q = 2$ would give $(q + q')^2 = 0$, hence $q' = -q$, when q, q' are pure imaginary of norm 1 with $q' \neq -q$, while $q' = -q$ gives the same conjugation; the structure constants are rational, so one computation settles all of them, and solving the linear conditions on the sixteen coordinates of D, for a conjugation α, gives $\dim \mathrm{BSDer}_{\sigma_\alpha}(\mathbb{H})^\alpha = \dim \mathrm{Der}(\mathbb{H})^\alpha = 1$; for $\alpha = \mathrm{id}$ the same computation gives 3 for both, and the two classes agree by Theorem 3.4. So the conclusion of Theorem 4.1 holds at $\mathbb{H}$ for every involutive α, although $\mathbb{H}$ is not commutative. The same computation gives $\dim \mathrm{BSDer}_{\sigma_\alpha}(\mathbb{H}) = 1 < 3 = \dim \mathrm{Der}(\mathbb{H})$ for $\alpha \neq \mathrm{id}$: at $\mathbb{H}$ the derivations lie in the twisted class only in so far as they commute with α. For a general reduced algebra with 2 invertible and α an involutive automorphism, the argument of Theorem 4.1 reduces the question to the skew part. Let $E = E_D$ and $E_\mathrm{a}(x, y) = E(x, y) - E(y, x)$, twice the antisymmetric part of Section 3. Then $E_\mathrm{a}(x, y) = D([x, y]) - [D(x), y] - [x, D(y)]$, so $E_\mathrm{a} = 0$ says that D is a derivation for the commutator bracket. Proposition 3.2(i), applied at (x, y) and at (y, x), gives $\alpha(E_\mathrm{a}(x, y)) = E_\mathrm{a}(x, y)$, so E_a takes values in A_+. Since α is multiplicative and D commutes with it, E maps $A_i \times A_j$ into A_{i+j}; since E is additive in each argument and $A = A_+ + A_-$, three cases remain. If $x \in A_+$ and $y \in A_-$ then $E(x, y)$ and $E(y, x)$ lie in A_-, so $E_\mathrm{a}(x, y) \in A_+ \cap A_- = 0$. If $x, y \in A_+$ then $x \cdot_{\sigma_\alpha} y = xy + yx$, so the restriction of D to A_+ is a Jordan derivation of A_+, a reduced, hence semiprime, and 2-torsion-free ring, and is therefore a derivation there by Theorem 3.4; hence $E(x, y) = 0$. So only $x, y \in A_-$ remains, where E is alternating with values in A_+: the values lie in A_+ by the parity of E, and $E(x, x) \in A_+ \cap A_- = 0$ because Proposition 3.2(i) at (x, x) puts $E(x, x)$ in A_-. If it vanishes there then $E_\mathrm{a} = 0$, the defect is symmetric, and the argument at the end of the proof of Theorem 4.1 makes D a derivation; whether it must vanish is what is missing, and it is not settled here.

The three conditions are in fact equivalent. Write (i) for $D \in \mathrm{Der}(A)$, (ii) for $E = 0$ on $A_+ \times A_-$, and (iii) for $E = 0$ on $A_- \times A_-$. A derivation has $E = 0$, so (i) implies (ii) and (iii). Assume (iii). Then E vanishes on $A_+ \times A_+$ by the argument of the case $x, y \in A_+$ above, and on $A_- \times A_-$ by hypothesis; on a mixed pair $E(x, y) = E(y, x)$, because both lie in A_- and their difference lies in A_+. So E is symmetric, and the argument of the proof of Theorem 4.1, which uses only the symmetry of E and not the commutativity of A, gives $E = 0$; so D is a derivation. Assume (ii). Then E vanishes on $A_+ \times A_+$ as above and on the mixed pairs by hypothesis, and on $A_- \times A_-$ it is alternating, hence antisymmetric, because $E(x, x)$ lies in A_+ by the parity of E and in A_- by Proposition 3.2(i) at (x, x). So E is antisymmetric, which makes D a Jordan derivation of A, and A is reduced, hence semiprime and 2-torsion-free, so the two forms of the Jordan condition agree by the linearization of Section 2 and D is a derivation by Theorem 3.4. Thus (i), (ii) and (iii) are equivalent, and a counterexample to the conclusion for reduced A has a nonzero defect on the mixed pairs as well as on $A_- \times A_-$; the example above, where E vanishes on the mixed pairs, escapes this because its algebra is not reduced.

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